

POLYTECHNIC UNIVERSITY OF BUCHAREST

PHYSICS

Magnetic field generated by Helmholtz coils analysis

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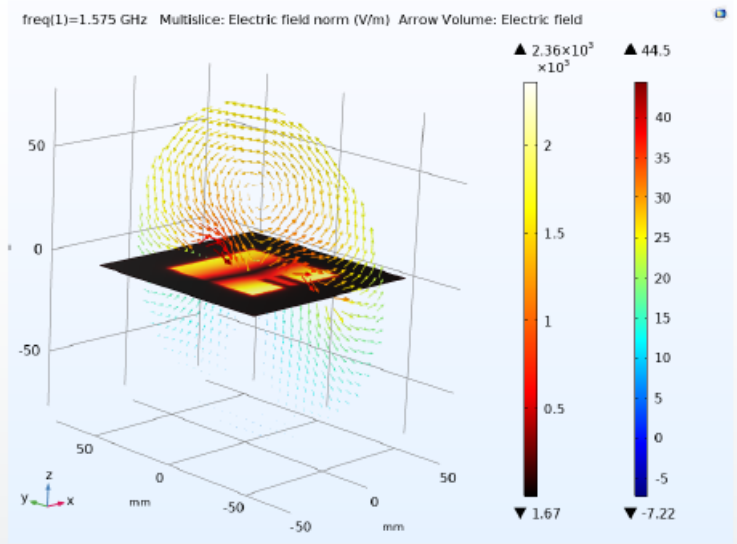
Abstract

Magnetic fields are an important physics concept now-days, thus we have to continue studying them. They are generated by a multitude of devices, one of them being coil-form devices. A type of coils would be Helmholtz coils, and in this paper work we are going to analyze the magnetic field generated by them, starting from scratch.

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0.1 Introduction



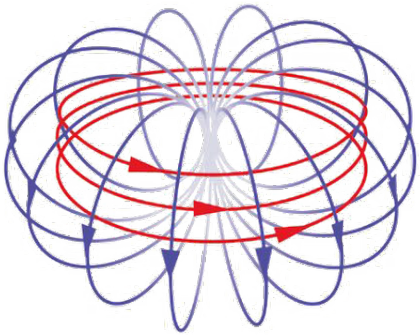
Magnetic fields are a fundamental part of our everyday lives. They are created by the movement of electric charges, and can be found in a wide range of natural and man-made sources. The Earth itself has a magnetic field, which is what allows compasses to point north. Electric power generators, motors, and speakers all make use of magnetic fields to function. Magnetic fields are also used in medical imaging, such as in MRI machines. Additionally, magnetic fields play an important role in the functioning of the Earth's atmosphere, including the aurorae seen in the polar regions. Overall, magnetic fields are an important and ubiquitous aspect of our physical world.

The interesting part about electromagnetic fields is that we, humans, can't see or feel it. That's why this phenomenon aroused such interest in our team. We wanted to learn more about electromagnetic fields so our complex project is based on that topic. The idea of this project is to try to create and visualize a map of an electromagnetic field. Because we can't see an EM field and simply draw it we tried to achieve "superpowers" by using science and technology because "Any sufficiently advanced technology is indistinguishable from magic.".

The tool for generating magnetic fields we used is the Helmholtz coil, which consists of two identical circular coils placed in parallel and separated by a distance equal to their radius. In this study, we aim to map the magnetic field generated by a Helmholtz coil using a variety of experimental techniques and to investigate the factors that affect the strength of the magnetic field.

0.2 Theory Fundamentals

What Does Electromagnetism Mean?

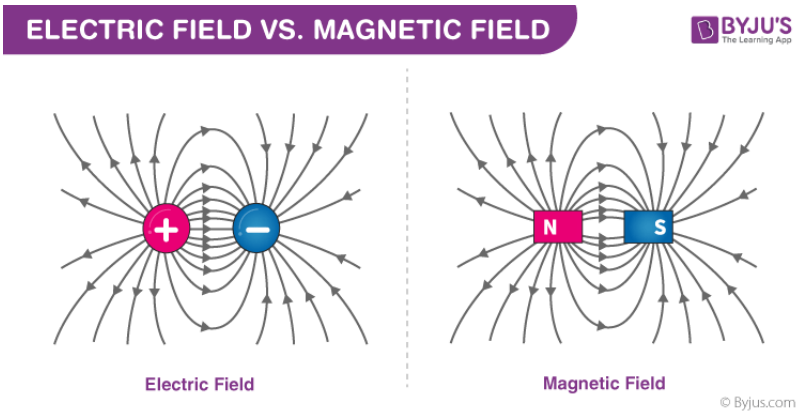


Electromagnetism is a branch of physics which deals with study of electromagnetic force. Being one of four fundamental interactions – along with gravitation, weak interaction and strong interaction – electromagnetic force happens between electrically charged particles. Discovered in the 19th century, electromagnetism has extensive usage in todays' physics.

Electricity and magnetism were long thought to be separate forces. In 1905 Albert Einstein's special theory of relativity established beyond a doubt that both are aspects of one common phenomenon. At a practical level, however, electric and magnetic forces behave quite differently and are described by different equations. Electric forces are produced by electric charges either at rest or in motion. Magnetic forces, on the other hand, are produced only by moving charges and act solely on charges in motion.

Electric and magnetic forces can be detected in regions called electric and magnetic fields. These fields are fundamental in nature and can exist in space far from the charge or current that generated them. Remarkably, electric fields can produce magnetic fields and vice versa, independent of any external charge. A changing magnetic field produces an electric field, as the English physicist Michael Faraday discovered in work that forms the basis of electric power generation. Conversely, a changing electric field produces a magnetic field, as the Scottish physicist James Clerk Maxwell deduced. The mathematical equations formulated by Maxwell incorporated light and wave phenomena into electromagnetism. He showed that electric and magnetic fields travel together through space as waves of electromagnetic radiation, with the changing fields mutually sustaining each other.

Maxwell's equations still provide a complete and elegant description of electromagnetism down to, but not including, the subatomic scale. The interpretation of his work, however, was broadened in the 20th century. Einstein's special relativity theory merged electric and magnetic fields into one common field and limited the velocity of all matter to the velocity of electromagnetic radiation.



Everyday modern life is pervaded by electromagnetic phenomena. When a lightbulb is switched on, a current flows through a thin filament in the bulb, and the current heats the filament to such a high temperature that it glows, illuminating its surroundings. Electric clocks and connections link simple devices of this kind into complex systems such as traffic lights that are timed and synchronized with the speed of vehicular flow. Radio and television sets receive information carried by electromagnetic waves traveling through space at the speed of light. To start an automobile, currents in an electric starter motor generate magnetic fields that rotate the motor shaft and drive engine pistons to compress an explosive mixture of gasoline and air; the spark initiating the combustion is an electric discharge, which makes up a momentary current flow.

A magnetic field is created whenever charge is in motion–either moving in space or spinning around itself. A charge moving in space is called a "current" (denoted by the symbol I) and is measured in coulombs/sec or amperes.

The strength of magnetic field is measured at a point in space (often called the field point). In the case of the Helmholtz coils, the field points of interest are located in the mid-plane between the two coils. As shown in the equation above, the strength of the magnetic field is dependent upon three quantities:

- the current I,
- the number of turns N in each coil,
- the radius a of the coil.

The total current in each coil is NI.

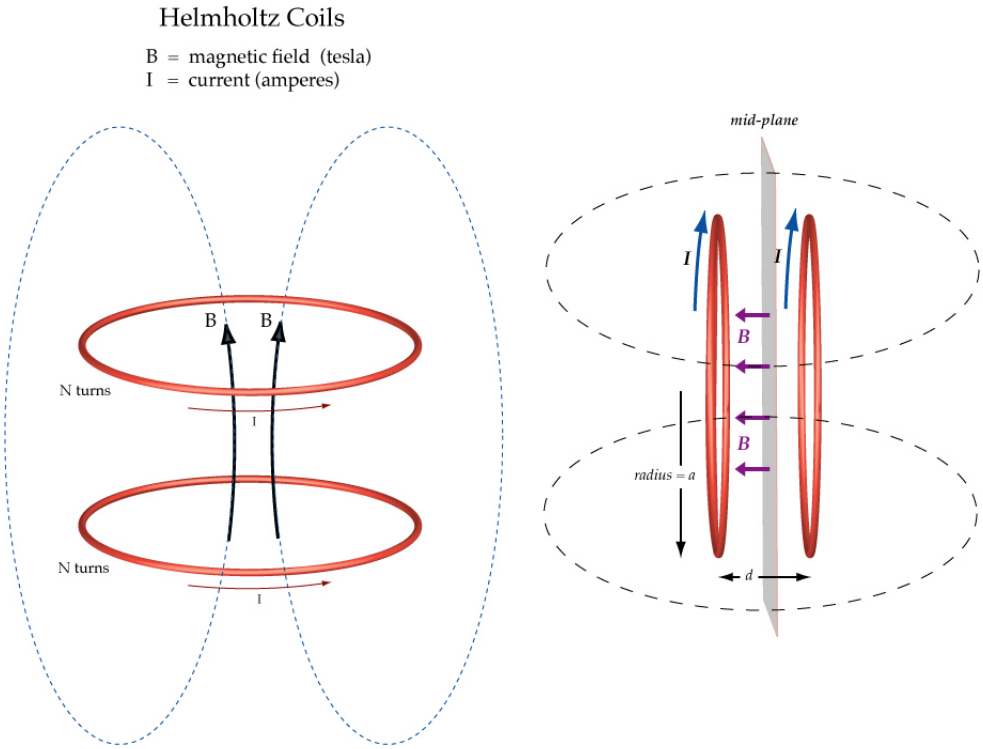
What are the electromagnetic coils?

An electromagnetic coil is an electrical conductor such as a wire in the shape of a coil (spiral or helix). Electromagnetic coils are used in electrical engineering, in applications where electric currents interact with magnetic fields, in devices such as electric motors, generators, inductors, electromagnets, transformers, and sensor coils. Either an electric current is passed through the wire of the coil to generate a magnetic field, or conversely, an external time-varying magnetic field through the interior of the coil generates an EMF (voltage) in the conductor.

A current through any conductor creates a circular magnetic field around the conductor due to Ampere's law. The advantage of using the coil shape is that it increases the strength of the magnetic field produced by a given current. The magnetic fields generated by the separate turns of wire all pass through the center of the coil and add (superpose) to produce a strong field there. The more turns of wire, the stronger the field produced. Conversely, a changing external magnetic flux induces a voltage in a conductor such as a wire, due to Faraday's law of induction. The induced voltage can be increased by winding the wire into a coil because the field lines intersect the circuit multiple times.

The direction of the magnetic field produced by a coil can be determined by the right hand grip rule. If the fingers of the right hand are wrapped around the magnetic core of a coil in the direction of conventional current through the wire, the thumb will point in the direction the magnetic field lines pass through the coil. The end of a magnetic core from which the field lines emerge is defined to be the North pole.

What are Helmholtz coils?
A pair of conducting circular coils each having N turns, each carrying a current I, separated by a distance equivalent to the radius of the circular loops, produce a homogeneous magnetic field B in the mid-plane between the two circular coils.



$$B = \frac{32\,\pi\,N\,I}{5\sqrt{5}\,a} \times 10^{-7} \text{ tesla}$$

a = radius of the coils a = separation between the coils a/2 = distance to the mid-plane B = magnetic field at the mid-plane

About the electromagnetic field in Helmholtz coils

Helmholtz coils are a type of electromagnetic apparatus used to create a region of nearly uniform magnetic field in a defined area. They are composed of two coils of wire wound around a common axis and separated by a small distance. The coils are typically square, circular, or cylindrical in shape, with each coil carrying the same current, but in opposite directions. The magnetic field created by Helmholtz coils is uniform within a certain volume and is approximately zero outside this volume.

The general purpose of Helmholtz coils is to create a uniform magnetic field in a defined area. This uniform field can be used for a variety of applications, including scientific experiments, medical imaging, radar, and industrial processes. By creating a uniform field, Helmholtz coils make it easier to measure and control the magnetic field, allowing for more precise and accurate measurements and results.

How uniform is the magnetic field?

The "claim to fame" for the Helmholtz coil is its very uniform magnetic field. However, the question that naturally arises , "how uniform, and over what distance?" To determine the uniformity of the magnetic field requires more than the above equation and involves some complicated integral calculus. However, we can generate some figures that graphically show how the magnetic field varies on the mid-plane as a function of r (the distance from the axis of symmetry passing between the centers of the two coils) and d the separation between the two coils.

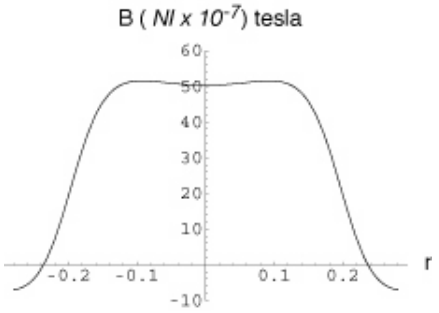


Figure 1: Coil spacing is "too close."d = 0.8 $\cdot a$

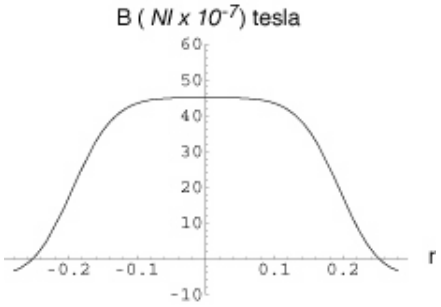


Figure 2: Coil spacing is "just right."; d = a

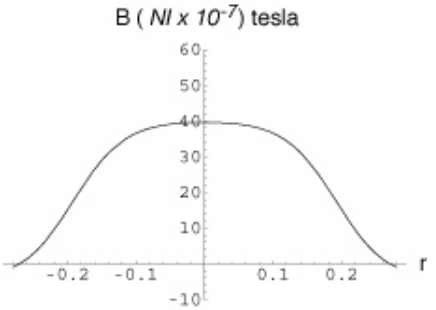


Figure 3: Coil spacing is "too far."; d = 1.2 $\cdot a$

In the above figures, we can see how uniform the magnetic field is on the mid-plane when the spacing d is too close, just right, and too far. The figures to the right are calculated for coils with a radius of 20 cm (i.e., a = 0.20 m).

The following integral was evaluated to produce the three curves above.

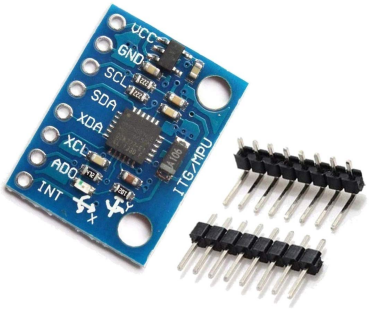
$$B_z(r) \;=\; N\,I\;\times 10^{-7}\int_0^{2\pi}\frac{2a(a-r\cos(\theta))\,d\theta}{\left(a^2+b^2+r^2-2ar\cos(\theta)\right)^{3/2}}$$

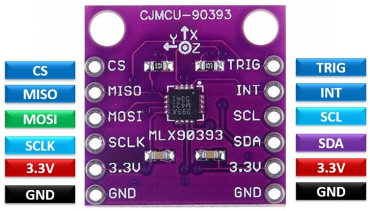
where the factor of "2" in the numerator takes into account the two coils, and b = d/2 is the distance from each coil to the mid-plane.

0.3 Tehnical Implementation

0.3.1 The plan

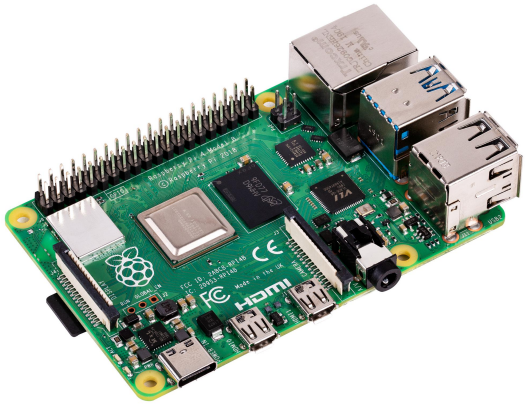
In order to map the intensity of the magnetic field at a given location, at any given point near the coils, two sensors were to be used: the MPU-6050 (used as an accelerometer) and the Melexis MLX90393 (used as an magnetometer). Both sensors can measure vector values on all three axis, so by processing the data and selecting the first point's position as a reference, the path of the device can be traced, along with the recorded magnetometer data.



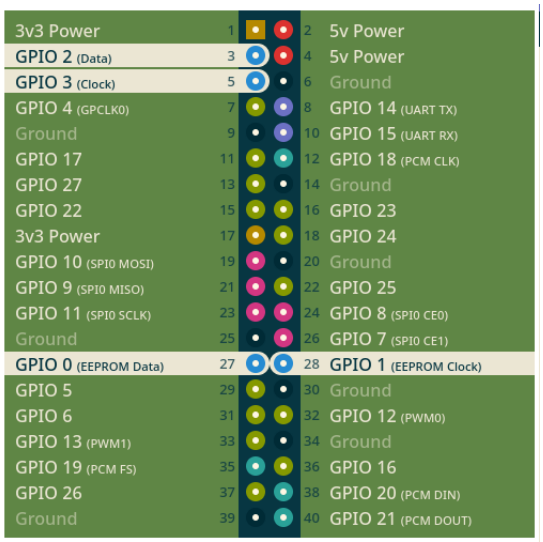


0.3.2 Details about the observing device

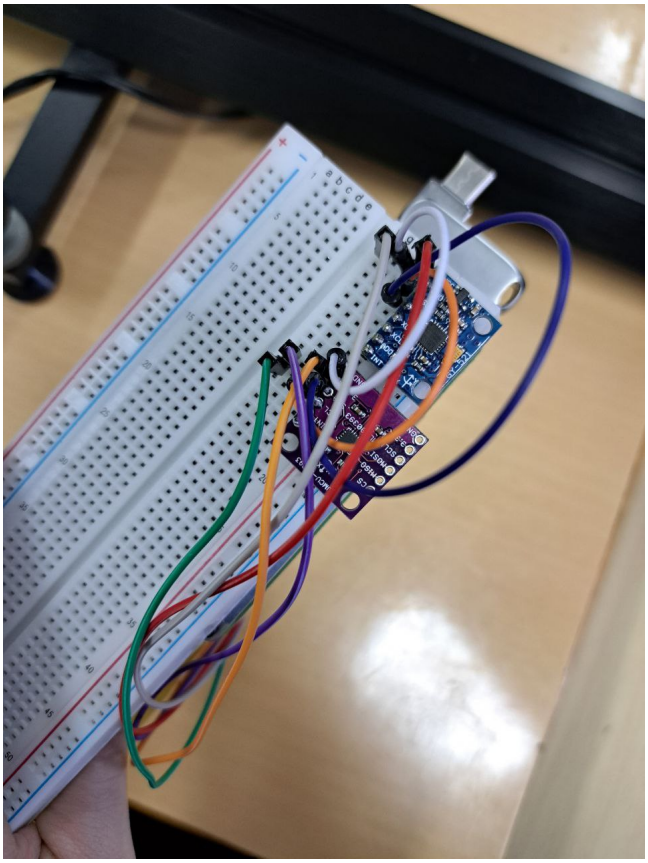
To make use of these sensors (which provide digital output), they both needed to be connected to the same development board, and placed near one another. A Raspberry Pi 4 was used for this purpose, which was attached to a breadboard along with the sensors. To start a test run, modify the code used or visualize the data, the Pi was ran in headless mode, and controlled via SSH by a nearby laptop.



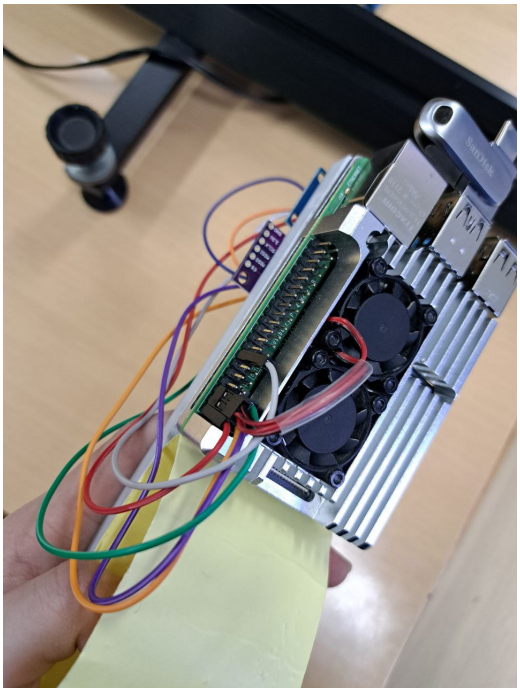
There were some drawbacks initially, but after some tinkering on the Pi, it was possible to connect and use both sensors on the i2c interface. The data endpoint of each sensor was connected in series with the GPIO2 (Data) pin on the Pi, while the clock of each sensor was connected in the same way to GPIO3 (Clock). The MLX was powered by the 3.3V supply, while the MPU was powered by the 5V supply, and all ground pins were connected.



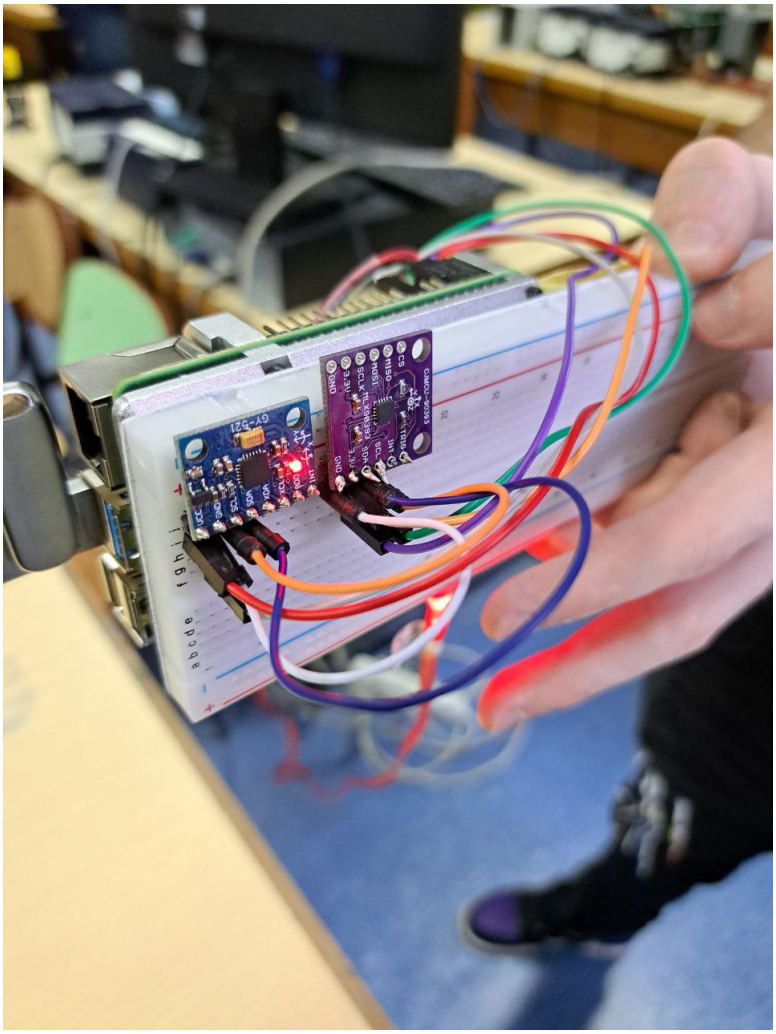
This was working well, as both sensors had different i2c addresses, so there was no corruption of data when connecting both their outputs in series.
The board with the sensors was the following:



The connection to the Raspberry Pi was the following:



The final device was looking like this :



0.3.3 Python program

In order to get data, we used Python 3.6 and the libraries for the sensors , more specifically: **import adafruit_mlx90393** and **mpu6050**. These libraries have modules supported for the sensors, that helped us process the data into values for the magnetic field and acceleration given a certain position. The code for the program will look like this as a result:

```
import time
import board
import adafruit_mlx90393
from mpu6050 import mpu6050

i2c = board.I2C()  # uses board.SCL and board.SDA
SENSOR = adafruit_mlx90393.MLX90393(i2c, gain=adafruit_mlx90393.GAIN_1X)
SENSOR_P = mpu6050(0x68)

count = 0

#SENSOR.self_test()
while True:
    MX, MY, MZ = SENSOR.magnetic
    count += 1
    print(MX)
    print(MY)
    print(MZ)
    accel_data = SENSOR_P.get_accel_data()
    print(accel_data['x'])
    print(accel_data['y'])
    print(accel_data['z'])
   # Display the status field if an error occured, etc.
    if SENSOR.last_status > adafruit_mlx90393.STATUS_OK:
        SENSOR.display_status()
    time.sleep(0.05)
```

As we can see we imported the libraries. With **SENSOR.magnetic** we got data for the magnetic field in vector format , and with **SENSOR_P** for the acceleration position, also in vector form. We then wrote data in a file for 1.8A and 2A using the command **main.py > 1.8.A** and **main.py > 2.0.A** in the terminal. As the program was running , we got the data by moving the device around the coils and it was live written in the respective files.



0.3.4 Error

First of all, there were errors to the software part of it because we didn't find the best modules at start, and then we got bugs with the version of python, which made us downgrade it in order for the modules to work - and we didn't know that - it was an obstacle because we lost time.

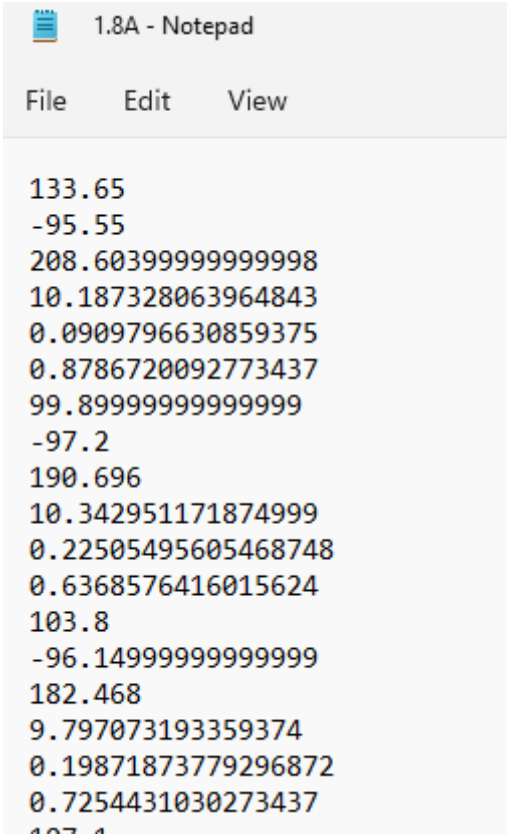
Second, we started with just the accelerometer and the device provided for the magnetic intensity was not adapted to the Raspberry Pi 6 , so we had to get a new magnetometer in order to get this specific, but important data.

Also, there is to mention that we had some bad attempts to get data, as it was not calibrated from the start and the sensors were not that sensitive, as ironic as it would be. And to add to that, we had to keep the "device" very still in order to get the data correctly for the last good attempt, which is a sign that the devices do not work that well when it comes to rotating it - it gave us data that we were unable to process, which made us do it more times , instead of just one time gathering the data and start to process it.

0.4 Data Analysis

0.4.1 Data pre-processing for 1.8A

After implementing the program and getting the data, the file with data will look like this :



As we can see above, we have a list of data , the first three lines represents the data for the magnetic field given the axes X, Y, and Z , and the next three for the acceleration on the same axes. And this goes on until the last data. So we imported it in our new python program for the analysis of the gathered values and to represent them :

```
In [1]: #Lucrare Complexa Fizica

In [2]: import pandas as pd
        import numpy as np

In [3]: df=pd.DataFrame()

In [4]: #read from file
        with open("1.8.A") as file:
            values=[line.rstrip() for line in file]

In [5]: valuesFinal=np.array(values)

In [6]: #create bidimensional array with the data
        k=0
        array=np.reshape(valuesFinal, (420,6 ),"C")

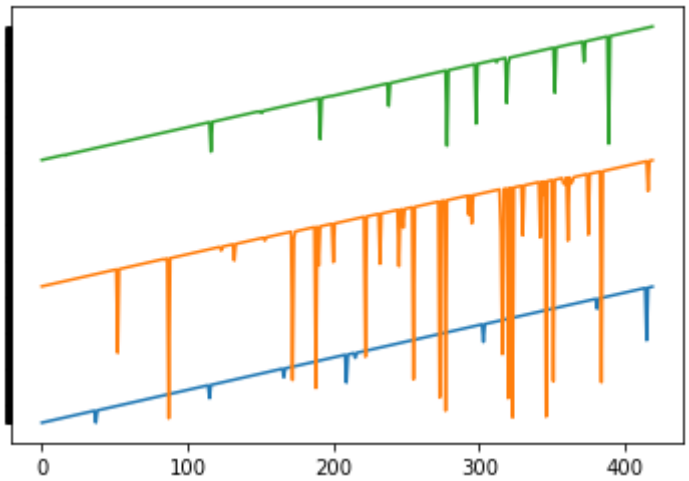
In [7]: #create a dataframe in order to easly represent the data
        df=pd.DataFrame(array, columns=["mag_x","mag_y","mag_z","pos_x","pos_y","pos_z"])
```

And the values will look like this:

	mag_x	mag_y	mag_z	pos_x	pos_y	pos_z
0	14.549999999999999	-260.84999999999997	97.042	9.562441430664062	-0.04548983154296875	1.9536685546875
1	39.9	-278.09999999999997	105.75399999999999	9.892841259765625	-0.13168109130859373	2.1691467041015624
2	60.3	-291.15	105.512	10.067617980957031	-0.09816226806640624	2.2002713256835937
3	88.64999999999999	-312.9	107.448	9.935936889648437	-0.062249243164062495	2.2481553588867187
4	120.89999999999999	-338.25	101.398	9.512163195800781	-0.011971008300781249	2.437297290039062
...	...	...	...	...	...	...
415	1284.3	30.599999999999998	234.73999999999998	10.187328063964843	-0.03591302490234375	2.252943762207031
416	939.3	31.2	356.70799999999997	10.237606298828124	-0.19632453613281248	2.2098481323242187
417	551.85	70.8	473.836	9.938331091308592	-0.09576806640624999	2.779668127441406
418	151.2	38.699999999999996	318.956	9.579200842285156	0.8786720092773437	3.1100679565429687
419	-103.5	-4.8	167.948	9.935936889648437	0.5482721801757812	1.5682020874023437

420 rows × 6 columns

If we would have to represent the magnetic field's intensity in vectorial form, we have this form:



As we can see, even in vector form , we have a different representation in dependence of the intensity. Now , in order to calculate the magnetic intensity we will use the formula :

$$Value = \sqrt{x^2 + y^2 + z^2} \tag{1}$$

```
array_mag=[0]*420
```

```
#Given the fact that we have the magnetic field data given in vector form
#we will use the magnitude to find the true value in the given point
for i in range(0,420):
    array_mag[i]=np.sqrt(float(array[i][0])**2+float(array[i][1])**2+float(array[i][2])**2)
```

And we got the following values :

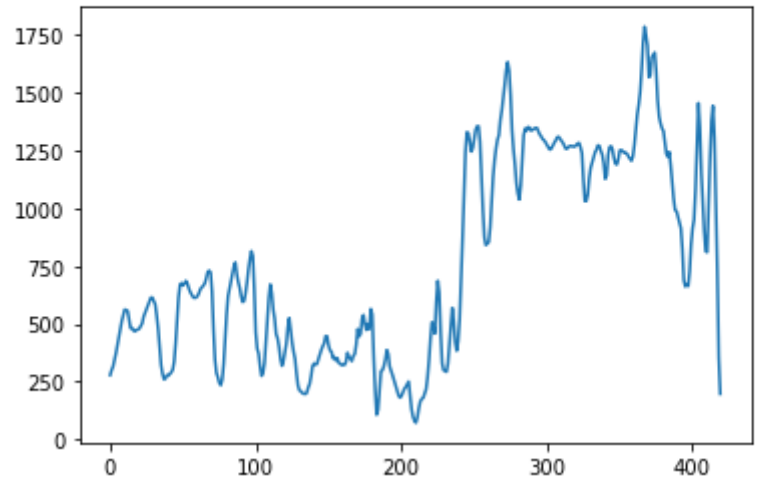
	mag_x	mag_y	mag_z	pos_x	pos_y	pos_z	magnetic_magnitude
0	14.549999999999999	-260.84999999999997	97.042	9.562441430664062	-0.04548983154296875	1.9536685546875	278.696205
1	39.9	-278.09999999999997	105.75399999999999	9.892841259765625	-0.13168109130859373	2.1691467041015624	300.192486
2	60.3	-291.15	105.512	10.067617980957031	-0.09816226806640624	2.2002713256835937	315.495158
3	88.64999999999999	-312.9	107.448	9.935936889648437	-0.062249243164062495	2.2481553588867187	342.505920
4	120.89999999999999	-338.25	101.398	9.512163195800781	-0.011971008300781249	2.437297290039062	373.244460
...	...	...	...	...	...	...	...
415	1284.3	30.599999999999998	234.73999999999998	10.187328063964843	-0.03591302490234375	2.252943762207031	1305.934806
416	939.3	31.2	356.70799999999997	10.237606298828124	-0.19632453613281248	2.2098481323242187	1005.235558
417	551.85	70.8	473.836	9.938331091308592	-0.09576806640624999	2.779668127441406	730.802037
418	151.2	38.699999999999996	318.956	9.579200842285156	0.8786720092773437	3.1100679565429687	355.094438
419	-103.5	-4.8	167.948	9.935936889648437	0.5482721801757812	1.5682020874023437	197.336820

420 rows × 7 columns

After we mapped them to the coordinates of the acceleration and we plotted the intensity's distribution :

```
plt.plot(index,df['magnetic_magnitude'])
```

```
[<matplotlib.lines.Line2D at 0x1adc0e1a3d0>]
```



We already see a pattern here - the intensity for the measurements outside and inside the coils are smaller then those measured right next to coils, as our colleague measured them this way. After sorting the values, we can see that the smallest value was **72.348652** $*10^{-6}$ T and the greatest **1786.650478**$*10^{-6}$ T. Also, the low values between the greater ones are because of the switch to other areas of the coils.

0.4.2 Finding displacement

We use the trapezoid method of integration that interpolates data to find the velocity and then the displacement coordinates from the **scipy** library and also we detranded them in order to remove any side effects and noise from our data.
We set the time segment to 0.05, as this was the interval that each data was read, then we detrend the data, and after that we double integrate using the ttrapezoid method as stated above.

```
#set time segment
time=0.05
```

```
#create time array
time_array=np.linspace(0.0,420*time, 420)
```

```
import scipy.integrate
```

```
#calculate time differential
delta_t=np.mean(np.diff(time_array))
```

```
#convert acceleration to numpy array
x_acceleration=df["pos_x"].to_numpy()
y_acceleration=df["pos_y"].to_numpy()
z_acceleration=df["pos_z"].to_numpy()
```

```
from scipy import signal
```

```
#detrend values for acceleration
x_acceleration=signal.detrend(x_acceleration)
y_acceleration=signal.detrend(y_acceleration)
z_acceleration=signal.detrend(z_acceleration)
```

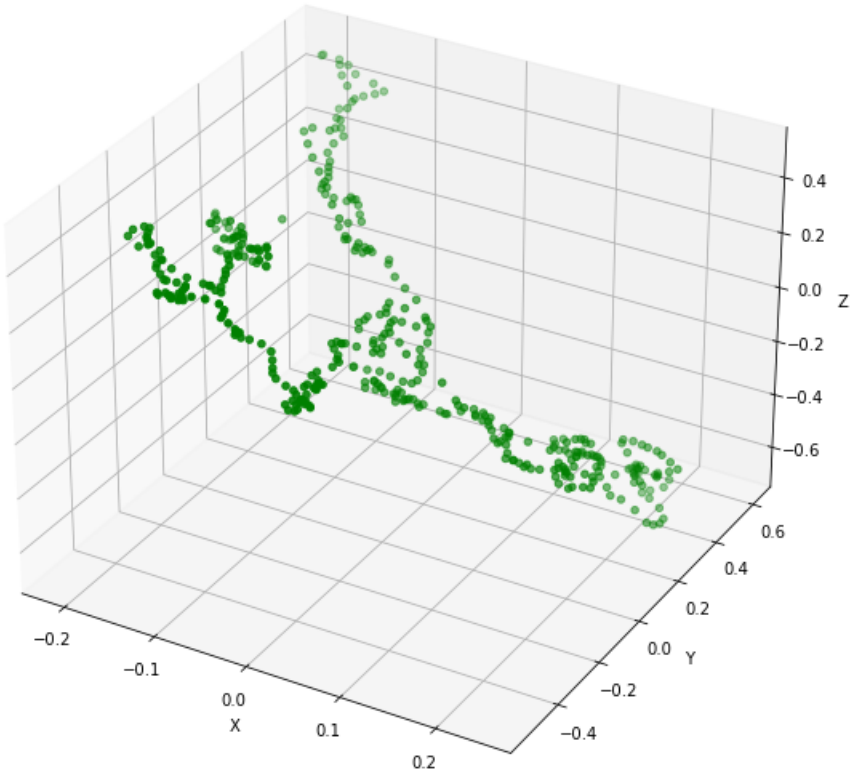
```
#find velocity given acceleration
x_speed=scipy.integrate.cumtrapz(x_acceleration, dx = delta_t, initial = 0)
y_speed=scipy.integrate.cumtrapz(y_acceleration, dx = delta_t, initial = 0)
z_speed=scipy.integrate.cumtrapz(z_acceleration, dx = delta_t, initial = 0)
```

```
#detrend velocity values
x_speed=signal.detrend(x_speed)
y_speed=signal.detrend(y_speed)
z_speed=signal.detrend(z_speed)
```

```
#find position given the velocity
x_position=scipy.integrate.cumtrapz(x_acceleration, dx = delta_t)
y_position=scipy.integrate.cumtrapz(y_acceleration, dx = delta_t)
z_position=scipy.integrate.cumtrapz(z_acceleration, dx = delta_t)
```

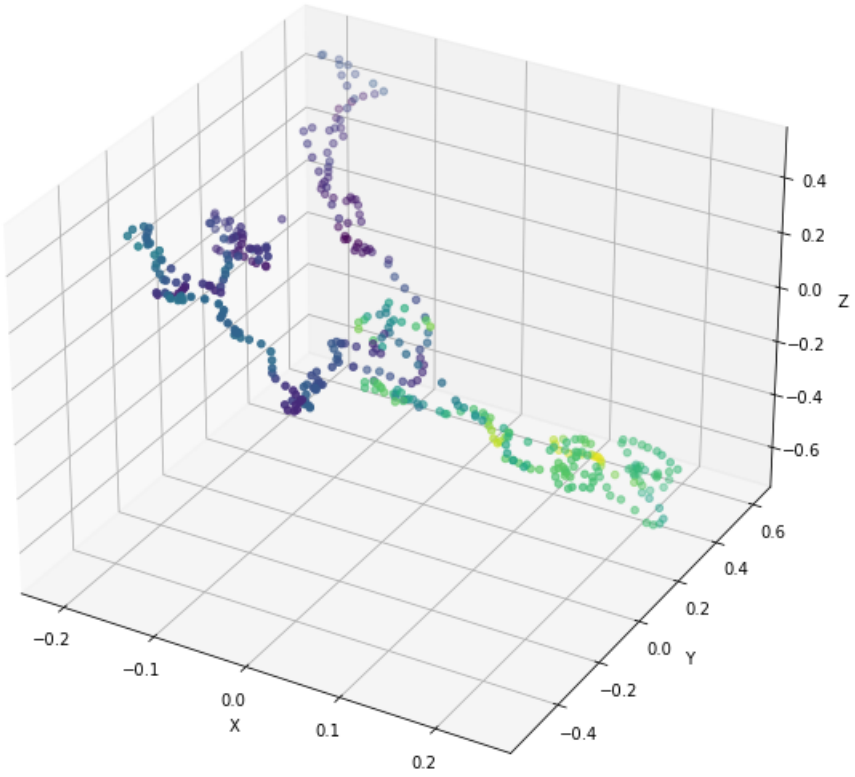
And after plotting the position of coordinates that depend on the acceleration we have something like this:

```
fig = plt.figure(figsize=(10,10))
ax = fig.add_subplot(111, projection='3d')
yg = ax.scatter(df2["x"], df2["y"], df2["z"],c="green")
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
plt.show()
```



Now we can do our final plot, which is represent every point according to the magnetic intensity we measured , in which the darker the colour , the greater the magnetic field's intensity:

```
fig = plt.figure(figsize=(10,10))
ax = fig.add_subplot(111, projection='3d')
yg = ax.scatter(df2["x"], df2["y"], df2["z"],c=df2["magnetic_input"] )
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
plt.show()
```



As we can observe the trajectory, the points for the places next to the coils are the darkest, and the rest are outside and inside the coils, or at least that's the conclusion that we came up with.

0.4.3 Data pre-processing for 2A

We will skip over the code as it is the same as above, except the data is different.

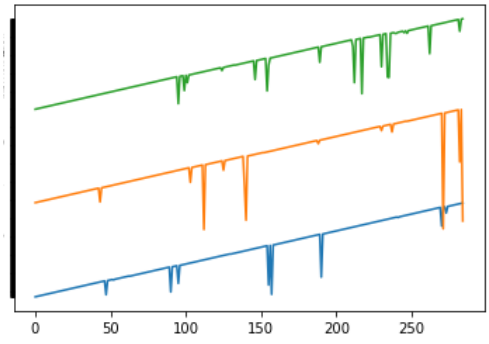
The data for 2A that was provided to the coils will look like this:

	mag_x	mag_y	mag_z	pos_x	pos_y	pos_z
0	1259.85	-242.39999999999998	390.58799999999997	9.861716638183593	-0.10773907470703124	2.2840683837890623
1	1299.0	-214.04999999999998	477.466	10.000580334472655	0.28730419921875	2.3822306518554686
2	1309.35	-183.0	533.3679999999999	9.897629663085937	0.08858546142578125	2.327164013671875
3	1333.35	-141.75	530.948	10.465055456542968	0.1292868896484375	2.528276953125
4	1385.3999999999999	-88.8	518.848	9.651026892089844	0.06943184814453125	2.6527754394531247
...	...	...	...	...	...	...
280	-334.5	-44.85	-182.226	10.12507882080078	0.5195417602539062	2.398990063476562
281	-199.35	-27.3	-180.29	9.892841259765625	0.22744915771484372	2.6048914062499997
282	-111.89999999999999	-36.15	-126.566	10.012551342773437	0.28012159423828126	2.4396914916992185
283	-54.75	-63.9	-28.798	9.931148486328125	0.17477672119140625	2.214636535644531
284	-26.4	-115.19999999999999	89.53999999999999	9.677363110351562	0.4213794921875	1.9201497314453124

285 rows × 6 columns

As we can observe, there are two times less data then for 1.8A, which is because we didn't moved the "device" as much as for the 1.8A, which will lead to quite a different output further on.

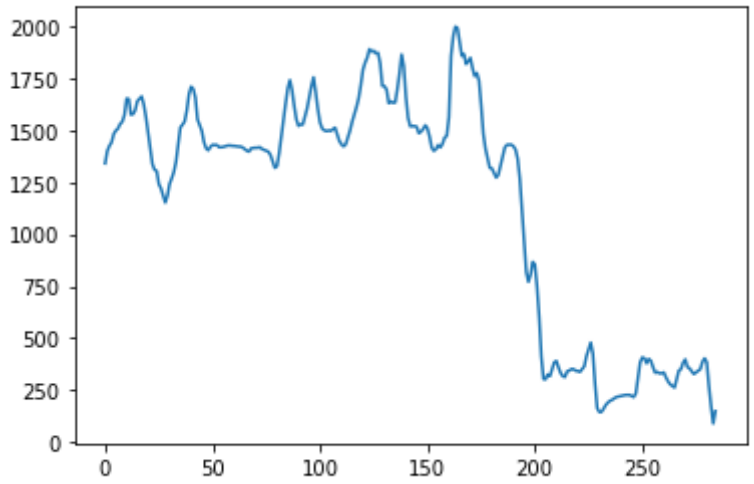
The vectorial representation for this case will look like this :



As for the magnetic field's intensity as final value we got the following values:

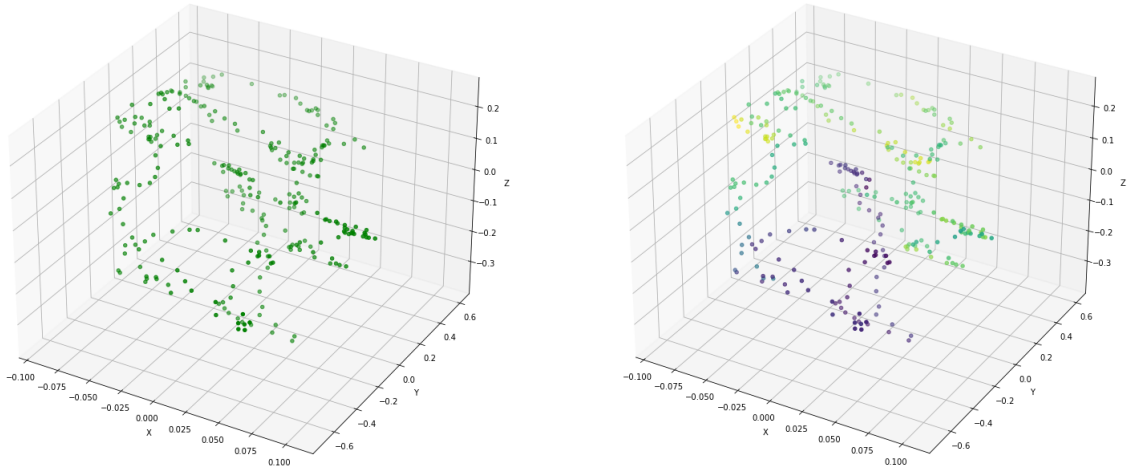
	mag_x	mag_y	mag_z	pos_x	pos_y	pos_z	magnetic_magnitude
0	14.549999999999999	-260.84999999999997	97.042	9.562441430664062	-0.04548983154296875	1.9536685546875	278.696205
1	39.9	-278.09999999999997	105.75399999999999	9.892841259765625	-0.13168109130859373	2.1691467041015624	300.192486
2	60.3	-291.15	105.512	10.067617980957031	-0.09816226806640624	2.2002713256835937	315.495158
3	88.64999999999999	-312.9	107.448	9.935936889648437	-0.062249243164062495	2.2481553588867187	342.505920
4	120.89999999999999	-338.25	101.398	9.512163195800781	-0.011971008300781249	2.437297290039062	373.244460
...	...	...	...	...	...	...	...
415	1284.3	30.599999999999998	234.73999999999998	10.187328063964843	-0.03591302490234375	2.252943762207031	1305.934806
416	939.3	31.2	356.70799999999997	10.237606298828124	-0.19632453613281248	2.2098481323242187	1005.235558
417	551.85	70.8	473.836	9.938331091308592	-0.09576806640624999	2.779668127441406	730.802037
418	151.2	38.699999999999996	318.956	9.579200842285156	0.8786720092773437	3.1100679565429687	355.094438
419	-103.5	-4.8	167.948	9.935936889648437	0.5482721801757812	1.5682020874023437	197.336820

420 rows × 7 columns



After sorting the values, we observed that the smallest value was **88.93872780740682** $*10^{-6}$ T and the greatest **1999.086332**$*10^{-6}$ T, which we can clearly tie to greater intensity input.

And last, but not least , the final plot for displacement and the mapped magnetic field's intensity to the displacement :



Here , we cannot observe as good as for the first case the trajectory because we got less data - thus much less to process. But still, we can see that we got more measurements for outside the coils , which are represented with lighter colours.

Overall, as a conclusion to this part , we can say that even the data for the first case (1.8A) was not enough as for a better result we should have had more than 1000 points in our opinion. even so, it was the first time making such an experiment and we didn't know how much of a data we are gonna need. Also, greater current input leads to greater magnetic field values.

0.5 Experience Records

0.5.1 Week 1:

In the first week we tried to figure out what were the exact steps we should follow in order to achieve our goal. We figured out what components we need for the measuring device. We compared the pros and the cons of the arduino board and a raspberry pi board and chose the last one.

0.5.2 Week 2:

In the second week we tried to connect the Helmholtz Coils to the voltage source but encountered a problem. The problem was that we tried multiple voltage sources but none of the would work properly. After the we focused on the raspberry pi and tried to connect the accelerometer to the board.

0.5.3 Week 3:

In the third week we found out that our magnetometer isn't compatible with our raspberry pi, thus we ordered another one that will fit our needs. Also we thought about how to collect the raw data and how to extract valuable information from it.

0.5.4 Week 4:

The fourth week was a bit unsuccessful for our team. We took the first measurements and created the data set. Unfortunately, the calibration algorithm wasn't working as we expected.

0.5.5 Week 5:

In the fifth week we took the measurements one more time. This time it was relatively successful. Also we started to work on the paper and delegated the responsibilities to each member of the team.

0.6 Conclusion

This project imposed a certain difficulty on us due to the lack of documentation, pushing us to look for more information about all the equipment used. As a result of doing this work, we learned some knowledge about the magnetic/electric and electromagnetic field, about Helmholtz coils and how they influence the value of the magnetic field's intensity.

Overall, it was an interesting experience, given the fact that we did not have a structure or an algorithm that we would follow, but instead, we had to create our own for this experiment.

For the hardware part of it, we can state that the lack of a well determined structure made us loose time because we had to switch to different sensors and get wrong data until we found a good sensor and a good way to measure it, as we had difficulties in keeping the "device" stable and we had to weld some pieces (with the help of assistants).

For the software part - there were errors in getting the "right" output given the fact that we did not think about getting more data and thought that it is sufficient the data that we collected. Also we had troubles getting the position coordinates and we stuck on it much more time than intended until we found an article about the trapezoid method.

Finally, we had some obstacles on the way, and we are certain we could have made much more research on this topic, but time is a limit itself and so, this is our final try concerning this topic.

0.7 References

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